# BRIAN MUMO MUSYOKI

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## PROFESSIONAL SUMMARY

Electrical and Electronics Engineering graduate with practical, project-based experience spanning IP networking, Wi-Fi, GPON/WISP architecture, network automation, Python/software development, AI/ML and electrical power systems. Rather than a purely academic exposure to networking, I have built and deployed working systems: my strongest technical project is an end-to-end UniFi-based captive portal and Wi-Fi billing platform that integrates access-point infrastructure, a controller API, a PostgreSQL-backed Flask application and M-Pesa payment processing to automate client network authorization. This is supported by GPON/WISP network design, Cisco/MikroTik-oriented simulation, and an electrical engineering foundation in power systems and ICT infrastructure. I am now looking to build on this practical foundation within a carrier-grade IP networking environment.

## WEBSITES, PORTFOLIOS, PROFILES

LinkedIn: https://www.linkedin.com/in/mumo-musyoki-b43288323

Project Portfolio Website: https://prototype.kesltd.co.ke/plans

IP Showcase website: https://showcase.kesltd.co.ke/

## ACCOMPLISHMENTS

**1. Telecommunications & IP Networking**

I have developed practical experience in IP networking, Wi-Fi infrastructure, GPON/WISP architectures, fibre access networks and network service delivery. I designed and developed a UniFi-based captive portal and billing platform for a university/student-hostel environment, integrating UniFi Access Points, EdgeSwitch, the UniFi Controller API, Python/Flask, PostgreSQL, REST APIs and PayHero/M-Pesa STK Push, so that a successful payment triggers time-based network authorization. This involved client/session management, controller API integration, database design, webhooks and Linux deployment. I have also used Cisco Packet Tracer and MikroTik-oriented networking concepts to model routing, switching, access networks and service delivery, and have developed academic and simulation-based knowledge of OSPF, BGP, MPLS and VPN fundamentals and service-provider architecture — knowledge I am actively working to deepen toward production-level competency.

**2. ICT, Software & AI**

I have developed software solutions using Python, Flask, PostgreSQL, SQL and REST APIs, including applications that integrate directly with physical network infrastructure such as access points and controllers. I have also built machine-learning and computer-vision workflows in Python and PyTorch, covering landmark detection, hand/body movement processing and skeletal representation. Alongside this I have worked with APIs, databases, web applications, automation, Linux/Ubuntu and Git/GitHub — a practical grounding in the software-driven, increasingly automated and data-intensive nature of modern telecommunications networks.

**3. Power Systems & Electrical Engineering**

I have practical experience in low-voltage electrical distribution, three-phase systems, electrical protection, solar PV, hybrid inverter and battery systems, EV charging infrastructure, generator systems and electrical testing and commissioning, using AutoCAD Electrical, AutoCAD, PVsyst and EasyPower. My work has included BOQ preparation, technical drawing review, site inspections, snagging and engineering documentation. This foundation supports my networking focus directly: reliable telecom infrastructure depends on consistent power availability, backup systems, electrical protection and equipment reliability, particularly at remote or access-network sites.

## TECHNICAL CAPABILITIES

**IP & Networking**

TCP/IP, Ethernet, IPv4 addressing, subnetting, routing fundamentals, switching, VLAN concepts, DHCP/DNS, Wi-Fi, network access control, client/session management, network troubleshooting, Cisco Packet Tracer, UniFi/UniFi Controller, EdgeSwitch, MikroTik concepts, GPON/WISP architecture, OLT and passive optical network concepts.

**IP / Service Provider Knowledge (fundamentals & simulation)**

OSPF fundamentals, BGP fundamentals, MPLS concepts, VPN concepts, service-provider architecture, IP/MPLS concepts, VPRN/VPLS concepts, Segment Routing concepts — developing knowledge from academic study and simulation, not production deployment.

**Software & Automation**

Python, Flask, REST APIs, PostgreSQL, SQL, Linux/Ubuntu, Git/GitHub, web application development, database-backed applications, API integration, webhooks, network automation.

**AI / ML**

Machine learning fundamentals, computer vision, PyTorch, landmark detection, skeletal representation, image/data processing, AI-enabled automation.

**Engineering**

AutoCAD Electrical, AutoCAD, PVsyst, EasyPower, low-voltage design, solar PV design, hybrid power systems, electrical protection, EV charging infrastructure, CCTV, access control, fire alarm, MEP coordination, BOQ preparation, technical reporting, testing and commissioning.

## PROFESSIONAL EXPERIENCE

**Project Manager — Arschan Associates Ltd, Nairobi, Kenya**

*May 2026 – Current*

- Coordinate electrical, ICT, CCTV, fire alarm, solar and MEP activities across active project sites.
- Review technical drawings and BOQs, and carry out site inspections to identify discrepancies between design documentation and site conditions.
- Manage snagging, quality control and technical coordination through to project close-out, including testing and commissioning.
- Track project follow-up and valuation documentation, and communicate progress and issues to stakeholders.

**Electrical Engineering Intern / Student Engineer — Jumba Engineering Services Ltd, Nairobi, Kenya**

*Jul 2021 – Current*

- Progressed from an administrative role into engineering while completing a BSc in Electrical and Electronics Engineering.
- Gained hands-on exposure to low-voltage systems, ICT infrastructure, CCTV, access control and time-attendance installations that connect network design to physical, on-site infrastructure.
- Contributed to MEP coordination, solar PV design, electrical documentation and BOQ preparation.
- Investigated faults, interpreted engineering drawings and coordinated technical work across live commercial-building projects.

## EDUCATION

**BSc Electrical and Electronics Engineering — University of Nairobi**

*August 2026*

Relevant coursework: Telecommunications, Digital Communications, Computer/Networking Concepts, Electronics, Control Systems, Electrical Power Systems, Engineering Computation.

The Nyeri High School — Secondary Education | Jan 2017 – Apr 2021

Lukenya Academy, Athi River — KCPE | May 2012 – Nov 2016

## LEADERSHIP

**Development Director, Rotaract Club of the University of Nairobi** — Jan 2025 / Incoming Rotary Year. Leading coordination, stakeholder engagement and professional development activities for the club, working across a team to plan and deliver initiatives.

## RELEVANT PROJECTS

**UniFi IP Network, Captive Portal & Wi-Fi Billing Platform**

Designed and developed an end-to-end Wi-Fi service platform for a university/student-hostel environment, built on UniFi Access Points, EdgeSwitch and the UniFi Controller. Developed the billing and authorization layer in Python/Flask with a PostgreSQL backend, REST APIs and PayHero/M-Pesa STK Push integration.

Workflow: a user connects to the Wi-Fi network and is redirected to the captive portal, selects a data plan, completes an M-Pesa payment, and a payment callback/webhook verifies the transaction and calls the UniFi Controller API to authorize the client for time-based internet access.

Handled client/session management, controller API integration, database design, webhook handling and Linux deployment — a working, if non-enterprise-scale, demonstration of software-driven network access control.

**GPON / WISP Telecommunications Network Architecture**

Designed and simulated an access-network architecture based on GPON and WISP principles, covering upstream connectivity, OLT placement, passive optical splitters, fibre distribution to subscribers, and Wi-Fi access points at the end-user side. This work developed my understanding of how physical access infrastructure connects to IP service delivery, from the access network through to the end user.

**Cisco / MikroTik Network Simulation**

Used Cisco Packet Tracer and MikroTik-oriented networking concepts to model IP networks, working through routing, switching, IP addressing, network topology, access-network design, service delivery and troubleshooting in a simulated environment.

**IP CCTV & ICT Infrastructure**

Worked on IP CCTV and ICT infrastructure deployments in commercial-building environments, including network-connected cameras, switches and structured cabling. This gave me a practical, physical-layer understanding of network-connected devices — connectivity, power requirements and fault isolation on real installations.

**AI / Machine Learning & Computer Vision**

Built Python and PyTorch-based machine learning and computer-vision workflows, including landmark detection and converting hand/body movement data into skeletal representations. This project developed my ability to work with data pipelines, automation and intelligent systems — capabilities increasingly relevant to modern, telemetry-driven telecommunications networks.

**Solar PV / Hybrid Power Systems**

Designed and modelled hybrid solar PV systems in PVsyst, covering PV string configuration, MPPT allocation, hybrid inverters, battery systems, grid interaction and load management. This is directly relevant to telecom infrastructure, where reliable power, backup systems and energy resilience are essential, particularly at remote sites.

**EV Charging & Electrical Infrastructure**

Worked on EV charging electrical design, including inverter integration, AC distribution, protection, surge protection (SPD) and APFC-based load management.

## CERTIFICATIONS

MATLAB Machine Learning Certified

MATLAB Onramp Certified

COMSOL Certified – IEEE MEMS Accelerometers

LUTRON Vive Certified

LUTRON RA2 Select Certified | Hikvision CCTV Installation, Design & Cost Estimation Certified | Hikvision Time Attendance & Access Control Certified

Centonomy Financial Management Certified

## LANGUAGES

English — Native/Bilingual | Kiswahili — Native/Bilingual | German — Limited Working Proficiency | French — Limited Working Proficiency

## CAREER TARGET FOR THE POSITION :

Seeking an IP Engineer opportunity with Nokia where I can apply my practical foundation in IP networking, telecommunications, software and engineering infrastructure. I want to deepen my expertise in carrier-grade IP routing, network automation and service-provider technologies, building from fundamentals toward production-level competency. My long-term goal is to combine IP networking with software, automation and AI to contribute to scalable, resilient and increasingly intelligent telecommunications infrastructure.